Centralizer extensions of projective-line maniplexes

human_compression project — private author draft

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1. Results and dependencies

A string marking of a group G is a generating ordered tuple $(r_0, \dots, r_{n-1})$ of distinct nonidentity involutions such that $r_i r_j = r_j r_i$ whenever $|i-j| > 1$. We do not impose the intersection property. A group with such a marking is often called an sggi. We write $G_I = \langle r_i : i \in I \rangle$ and $G_i = \langle r_j : j \neq i \rangle$. Its type is the sequence of adjacent product orders. Put $L(q) = \mathrm{PSL}_2(q)$ and $P(q) = \mathrm{PGL}_2(q)$.

This paper recovers the principal results of Leemans–Toledo [LT] from the common projective-line theory [E]. The change of category requires a proof: [E]'s classification assumes the intersection property, whereas a string marking need not have it. Sections 3–4 supply this interface. We then recover the source's two infinite constructions. A third infinite construction corrects its introduction and concluding face-count claim. The exact edition under consideration is the eleven-page author manuscript dated 15 May 2023, bound in the accompanying source ledger.

Theorem 1 (rank and local groups; Leemans–Toledo). If $L(q)$ has a string marking of rank at least four, then q is odd, the rank is four, and each end group G_0, G_3 is isomorphic to S_4, A_5, or $L(q)$. If both end groups are proper, there is exactly one marking up to isomorphism for $q = 11$, of type $\{3,5,3\}$, and exactly one for $q = 19$, of type $\{5,3,5\}$. These are the 11-cell and 57-cell; both satisfy IP.

Theorem 2 (extensions and the two source families).

1. Let q be odd, choose $\alpha \in \{1,-1\}$ with $q \equiv \alpha \pmod 4$, and put $N = (q-\alpha)/2$. Every rank-three string marking of $L(q)$ whose first adjacent order is an odd divisor of N extends, with the same group, to rank four.
2. For every prime $p \equiv 1 \pmod{12}$ there is such a marking with both end groups equal to $L(p)$. Its first three generators have type $\{3,p\}$.
3. For every prime $p \equiv \pm 1 \pmod{20}$, $p \neq 19$, there is such a marking with one end group A_5 and the other $L(p)$.

Both families in parts 2–3 are infinite and nonorientable. These are [LT, Theorem 4.4, Lemma 4.5, Proposition 4.7 and their corollaries].

Theorem 3 (the omitted symmetric-end family). For every prime $p \equiv \pm 1 \pmod{24}$ there is a rank-four string marking of $L(p)$ with facet group S_4 of type $\{3,4\}$ and vertex group $L(p)$. Consequently the associated regular maniplex has one vertex and $|L(p)|/24$ facets. There are infinitely many such examples.

Theorem 3 is a correction to the restricted face-count statement in the source's introduction and

conclusion. It is consistent with its correctly stated Theorem 3.10 and Remark 4.1, which allow S_4. No priority claim or new open-question credit is attached to the correction. Section 8 gives small exact witnesses for it and for two other literal source errors. The final PGL conjecture in [LT, §5] is treated in a separate packet; it is not silently included in the recovery of the above known results.

We retain the full common paper [E], including its classical Dickson subgroup input and exact finite-amalgam program. The parts used here are the characteristic-two involution centralizer, the centralizer descent for a nonabelian dihedral group, the six exceptional local markings and the ten universal presentations. We additionally retain the usual cyclic tori and their normalizers, and the maximal A_5 and S_4 cases of Dickson's theorem as printed in Giudici [G, Theorem 2.2(6),(8), p.4]. Dirichlet's theorem supplies infinitude of primes in the stated progressions. These outside theorems are not reproved or formalized here.

2. Marked groups and regular maniplexes

We recall the group–object correspondence so that the category change is explicit. Given a string marking, let the flags be the elements of G and let the i-adjacency be left multiplication by r_i. Each adjacency is fixed-point-free and involutory. The product of two distinct adjacencies is fixed-point-free because $r_i r_j \neq 1$. Nonconsecutive adjacencies commute, and connectedness is precisely generation of G. Thus these data satisfy the definition of a maniplex in [LT, §2].

A permutation of the flags preserves every adjacency exactly when it commutes with every left multiplication by a generator, hence with the left regular action of G. Such a permutation is right multiplication by its value at the identity: if $f(1) = h$, then $f(g) = gh$. Conversely every right multiplication commutes with the left action. The full colour-preserving automorphism group is therefore the right regular group, isomorphic to G by sending h to multiplication on the right by h^{-1}, and the maniplex is regular. This establishes the full automorphism group, rather than merely a flag-transitive subgroup.

Conversely, in a regular maniplex choose a base flag and the unique automorphism taking it to each adjacent flag. Automorphisms commute with the adjacency permutations. Moving the base flag along a coloured path therefore expresses its terminal automorphism as the corresponding reversed product of these distinguished automorphisms. They generate, are distinct nonidentity involutions, and commute at nonconsecutive positions. Changing the base flag conjugates the marking. This gives the claimed correspondence up to marking-preserving isomorphism.

An i-face is an orbit of the subgroup on every colour except i. For our regular model its number is $[G : G_i]$. Reversing the marking interchanges vertices and facets and gives the dual maniplex. In particular a reversal automorphism of the marked group gives a duality. The even-length word subgroup has index at most two. If G is a nonabelian simple group it is all of G: it is normal, and the quotient is generated by the common image of the involutions, of order at most two. This is exactly the source's nonorientability criterion.

3. Centralizer bounds without the intersection property

We record precisely the two results borrowed from the umbrella. In characteristic two the centralizer of a nonidentity involution in $P(q)$ is abelian [E, Lemma 3]. In odd characteristic, if a, b are involutions and $|ab| > 2$, their common centralizer in $P(K)$ has at most two elements. It is trivial in the unipotent case; otherwise its unique nonidentity element is represented by

$$A - \tfrac{1}{2}\operatorname{tr}(A)I, \tag{1}$$

where A represents ab. If a, b are defined over a subfield k, this whole centralizer lies in $P(k)$ [E, Theorem 5]. Neither centralizer statement uses IP. Their complete proofs remain in [E].

Lemma 4 (a string endpoint). In a centre-free group with a string marking, r_0 and r_1 do not commute.

Proof. Otherwise r_0 commutes with r_1 and, by the string relations, with all remaining generators. It then commutes with every word in the generators, so is central and hence trivial, a contradiction. □

Lemma 5 (rank obstruction). Suppose a centre-free group has the property that any two noncommuting involutions have at most one nonidentity common centralizer element. Every string marking of that group has rank at most four.

Proof. At rank at least five, Lemma 4 makes r_0, r_1 noncommuting. Both r_3 and r_4 centralize them. The hypothesis forces $r_3 = r_4$, contradicting distinctness of the marking. □

In characteristic two, a rank-four marking of $L(q)$ is also impossible: r_0, r_1 both centralize r_3, whose centralizer is abelian. This contradicts Lemma 4. For $q \geq 4$, the group $L(q)$ is nonabelian simple and centre-free, so these arguments apply. For $q = 2$, there are only three involutions in S_3; for $q = 3$, the involutions of A_4 generate only its four-group. These dispose of the small groups independently. We have proved the entire rank assertion of Theorem 1, including even characteristic, without importing the umbrella's C-group rank theorem.

For the remaining discussion q is odd and the rank is four. Every adjacent product has order at least three. An order-one product would repeat a marker. If an adjacent product has order two, divide the marking at that edge. The nontrivial subgroups generated on its two sides centralize one another and generate G. Each is therefore normal. Simplicity makes either one equal to G, whereupon the other lies in the trivial centre, a contradiction. This argument does not assume that the two sides have trivial intersection in advance.

4. Proper ends and the finite interface

Let $H = \langle r_0, r_1, r_2 \rangle$ be proper in $G = L(q)$ and put $D = \langle r_0, r_1 \rangle$. The first adjacent order exceeds two, so D is nonabelian. We apply the complete subgroup-type input of [E, §2].

A cyclic or elementary abelian group cannot contain D. In A_4 all involutions lie in a four-group, so they cannot generate A_4. In an odd-characteristic affine group an involution has multiplier -1 and form $x \mapsto -x + v$. Two such transformations commute precisely when their translation parameters agree. The distinct commuting endpoints r_0, r_2 therefore exclude the affine alternative, including its unipotent dihedral subgroups.

Suppose H is dihedral. Since r_0, r_2 are distinct commuting involutions, H has even rotation order and has a central involution z. Indeed, an odd dihedral group has only reflections, and two reflections commute only when they are equal. The involution z centralizes D. So does r_3. The common-centralizer bound forces $r_3 = z \in H$, contradicting $G = \langle H, r_3 \rangle$ and properness. This replaces both geometric and algebraic proofs of [LT, Lemma 3.9]; no assertion that every involution centralizer is maximal is needed.

If H is a subfield group, the retained projective form of Dickson's theorem conjugates it into $P(k)$ for a proper subfield $k \subsetneq \mathbb{F}_q$. Theorem 5 of [E] places r_3 in the same $P(k)$, since it centralizes D. Hence $G \subseteq P(k)$. Write $s = |k|$. Since $q \geq s^2$,

$$|G| = q(q^2-1)/2 > s(s^2-1) = |P(k)|, \tag{2}$$

which is impossible. This handles both $L(k)$ and $P(k)$ subgroups, including the small subfields; it uses their natural embeddings, not only their abstract isomorphism types. We conclude that H is S_4 or A_5. Reversing the marking proves the same assertion for G_0. This proves [LT, Theorem 3.10].

The source's odd-end-label lemma also follows without IP. If $|r_0 r_1|$ were even, its half-power would be the central involution of D. The common-centralizer bound would identify it with r_3, again putting r_3 in the proper group H. Thus the first label is odd whenever H is proper. This recovers [LT, Lemma 3.5] with its actual hypothesis. It does not assert that a marking with both ends equal to G has odd end labels.

Finite interface. Every rank-three string marking of S_4 or A_5 has IP. Up to automorphisms of the marked group its adjacent orders and Petrie order are exactly

$$\begin{array}{c|ccc} S_4 & (3,3,4) & (3,4,3) & (4,3,3) \\ A_5 & (3,5,5) & (5,3,5) & (5,5,3). \end{array} \tag{3}$$

Here the third entry is $|r_0 r_1 r_2|$. The new exhaustive program `phase258_local_sggi.g` enumerates all ordered triples of distinct involutions with commuting endpoints and tests generation. IP is an output test, never an admission filter. There are 72 such triples in S_4 and 360 in A_5; all pass the 64 subset-pair intersections. Full automorphism orbits give (3). An explicit dihedral string marking failing IP is a control against extending this assertion to arbitrary groups. The exact presentations for (3) and the ten compatible rank-four universals are already proved by [E, Lemmas 10–11] and its single bound GAP program. We reuse that calculation in full.

Suppose both end groups are proper. The local census now licenses the ten universal presentations, even though global IP is still unknown. Every possible G is a quotient preserving the four generators. Four rows are trivial. In the rows of orders 60 and 24 a stipulated end group is already the whole universal group, and remains the whole image in every quotient; this contradicts the required proper end. The order-96 row has no quotient $L(q)$ in odd characteristic: 60 does not divide 96, and for $q \geq 7$ the order is at least 168. The order-120 row is S_5, whose only nontrivial proper quotient is C_2; neither this nor S_5 is $L(q)$ for odd $q \geq 5$.

The remaining rows are the simple groups $L(11)$ and $L(19)$. Their only nontrivial quotients are themselves. The increasing odd-field order formula forces $q = 11$ or 19, and the bound universal markings already satisfy IP. Their reversal automorphisms give uniqueness under duality as well as ordered-marking uniqueness. This is the complete proof of [LT, Proposition 4.2], with the missing sggi-to-C interface made explicit. The standard local-map identifications give hemi-icosahedral facets for the 11-cell and hemi-dodecahedral facets for the 57-cell; their vertex figures are the dual local maps. The exact face counts, self-duality and markings are retained from the same complete table.

5. The half-turn extension theorem

Let q be odd and let $N = (q - \alpha)/2$ for $q \equiv \alpha \pmod 4$. Then N is even. A split or nonsplit cyclic torus of $L(q)$ of order N has a dihedral normalizer of order $2N$. For clarity, the split torus comes from diagonal projectivities of square eigenvalue ratio and has order $(q-1)/2$. The nonsplit torus comes from multiplication on $\mathbb{F}_{q^2}$ modulo scalar multiplications, followed by the determinant-square condition, and has order $(q+1)/2$. Inversion of the two eigenlines supplies the other normalizer coset. These standard torus facts do not assert maximality of the normalizer.

Suppose (r_0, r_1, r_2) generates $L(q)$, with distinct involutions, commuting endpoints, and $|r_0 r_1| = k$ an odd divisor of N. Necessarily $k \geq 3$. The element $r_0 r_1$ is semisimple and has two eigenlines over at most a quadratic field extension. The two possible PSL torus orders $(q-1)/2$ and $(q+1)/2$ are coprime; since $k > 1$ divides N, its torus has order N. Since $r_0(r_0 r_1)r_0 = (r_0 r_1)^{-1}$, both reflections preserve this unordered eigenline pair. Thus $D = \langle r_0, r_1 \rangle$ lies in the associated dihedral torus normalizer. Its cyclic torus has a unique involution z, which centralizes the normalizer and hence D.

The involution z is not in D: that dihedral group has odd rotation order and trivial centre, whereas z centralizes it. In particular $z \neq r_0, r_1$. It also differs from r_2, since z commutes with r_1 whereas $|r_1 r_2| \geq 3$. Therefore (r_0, r_1, r_2, z) is a string marking of rank four, and still generates the original group. By the group–maniplex correspondence its facets are the original map: there is one facet because the first three markers already generate G. This proves Theorem 2(1) and the complete scope of [LT, Theorem 4.4].

There is a useful local version. If a proper maximal subgroup $H \leq G$ has a rank-three string marking, its first adjacent order is an odd divisor of N, and $C_H(D) = 1$, the same torus involution z lies outside H. Maximality gives $\langle H, z \rangle = G$. The resulting rank-four marking has facet group H. The condition on the internal centralizer is essential: membership in the ambient torus alone does not prove that a new marker leaves the facet group.

6. The explicit family with two full ends

For a prime $p \equiv 1 \pmod{12}$, consider the projective classes of

$$R_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}, \quad R_2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad R_3 = \begin{pmatrix} 1 & -2 \\ 2 & -1 \end{pmatrix}. \tag{4}$$

The first three determinants are -1 and the last is 3. Both are squares modulo these primes: -1 is a square since $4 \mid p - 1$; if ζ has order three, then $(2\zeta + 1)^2 = -3$, and multiplication by a square root of -1 gives a square root of 3. Thus all four classes lie in $L(p)$. Their traces are zero and their determinants nonzero, giving nonidentity involutions. Direct multiplication shows that R_0, R_2 commute projectively, that $(R_0 R_1)^3$ is scalar and $R_0 R_1$ is nonscalar, and that

$$[R_1 R_2] = \left[\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \right]. \tag{5}$$

The latter element has exact order p. Its powers give all lower unipotent elementary matrices over $\mathbb{F}_p$. Conjugation by R_0 gives all upper ones. These generate $\mathrm{SL}_2(p)$: left multiplication by

lower and upper elementary matrices performs elementary row elimination. Explicitly, writing $u(t)$ and $v(t)$ for the upper and lower elementary matrices, put

$$h(t) = u(t)v(-t^{-1})u(t) = \begin{pmatrix} 0 & t \\ -t^{-1} & 0 \end{pmatrix}. \qquad h(t)h(-1) = \mathrm{diag}(t, t^{-1}).$$

The matrix $h(1)$ interchanges the two coordinate lines, and the displayed identity handles the last diagonal matrix after elimination. Passing to projective classes proves $\langle r_0, r_1, r_2 \rangle = L(p)$. This direct argument replaces a subgroup-type exclusion for this step.

For the fourth matrix, set $A = R_0R_1$. Then $2A - \mathrm{tr}(A)I = R_3$. Formula (1) shows that r_3 centralizes r_0, r_1. The first product has odd order three, so r_3 lies outside $\langle r_0, r_1 \rangle$. It differs from r_2 because the latter has product order p with r_1. Thus (4) is the extension of §5. The opposite end contains r_1r_2 of order $p \geq 13$. By Theorem 1 it is S_4, A_5 or $L(p)$; the first two groups have no element of that order. Both end groups are therefore $L(p)$, and the maniplex has one facet and one vertex.

These are the actual source generators, up to an explicit change of basis. Choose $a \in \mathbb{F}_p$ with $a^2 = -1$ and put $C = \mathrm{diag}(-a, 1)$. If S_0, S_1, S_2 denote the three matrices printed in [LT, Lemma 4.5], then the projective classes of CS_iC^{-1} are exactly the first three classes in (4). Indeed the three actual conjugated matrices are $-aR_0, aR_1, aR_2$. This identifies the construction at the level of its marked group, not just by order or type. The fourth generator is the unique nonidentity common centralizer element, so it agrees with the source's half-turn extension as well.

The prime progressions are nonempty and infinite by Dirichlet's theorem. Distinct primes give different group orders. Simplicity and §2 give nonorientability. This proves Theorem 2(2) and [LT, Corollary 4.6].

7. Alternating and symmetric proper ends

Let $p \equiv \pm 1 \pmod{20}$ be prime. Dickson's theorem gives a maximal A_5 subgroup H of $L(p)$ [G, Theorem 2.2(6)]. Give it the local marking $(5, 3, 5)$ from (3), corresponding to the hemi-dodecahedral map of type $\{5, 3\}$. Its first pair generates D_{10}. The internal centralizer $C_{A_5}(D_{10})$ is trivial: the centralizer of a 5-cycle in A_5 is its cyclic group of order five, and a reflection inverts that cycle, fixing only the identity in this centralizer.

For these primes, the torus order $N = (p - \alpha)/2$ is divisible by ten, where $p \equiv \alpha \pmod 4$. The construction of §5 gives a new involution outside H, hence a rank-four marking generating $L(p)$. If $p \neq 19$, then p is neither 11 nor 19. The two-proper-end classification in §4 forces the other end to be $L(p)$. Thus this is Class 2, with one vertex and $|L(p)|/60$ hemi-dodecahedral facets. For $p = 19$, the construction still gives a string marking, but the two-proper-end theorem no longer forces the opposite end to be full. The source excludes this value from its Class 2 assertion; no additional classification at this value is needed here. Infinitely many admissible primes remain, proving Theorem 2(3), [LT, Proposition 4.7] and Corollary 4.8.

Now let $p \equiv \pm 1 \pmod{24}$ be prime. Then $p \geq 23$, and Dickson's theorem gives a maximal S_4 subgroup H of $L(p)$ [G, Theorem 2.2(8)]. Use its local marking $(3, 4, 3)$ in (3). The first dihedral group is S_3, of order six. Its centralizer in S_4 is trivial: this S_3 fixes one point in the natural degree-four action and permutes the other three transitively, and an element centralizing that action must fix all four points. The relevant torus order $N = (p - \alpha)/2$ is divisible by twelve.

The construction of §5 therefore adjoins a centralizing involution outside H and generates $L(p)$ by maximality. Again the other end cannot be proper, since $p \notin \{11, 19\}$. This proves Theorem 3, including the complete all-prime range and the facet count. Dirichlet gives infinitude in both residue classes.

Consequently the correct classification of end-group sizes is:

Case	End groups, up to reversal	Vertices and facets
Both full	G, G	$1, 1$
One proper	G, S_4	$1, \lvert G\rvert/24$
One proper	G, A_5	$1, \lvert G\rvert/60$
Both proper	the 11-cell or 57-cell	$11, 11$ or $57, 57$

The middle two cases cannot be self-dual, since their vertex and facet counts differ. The first case can be self-dual; an exact example follows. We make no complete enumeration of its possible markings or dualities.

8. Literal source corrections and finite certificates

The new `phase258_local_sggi.g` supplies the precise category bridge used in §4. It also checks a counterexample to [LT, Lemma 3.3] as printed: in $L(5) \cong A_5$, an involution centralizer has order four and lies properly in a subgroup A_4 of order twelve. It is not maximal in A_5. The lemma also needs an odd-characteristic qualification for its displayed dihedral formula. Our proof uses the correct common centralizer theorem and torus normalizer, not this maximality assertion. The broad uniqueness clauses of source Lemma 3.4 are not retained inputs.

For the omitted symmetric-end case, the certificate gives four explicit permutations in the natural degree-24 action of $L(23)$. They are involutions, pairwise distinct, satisfy all string commutations and generate the group of order 6072. Their adjacent orders are $3, 4, 12$; the first end has order 24 and is S_4, while the other end is the full group. The maniplex therefore has one vertex and 253 facets. These facts directly contradict the introduction's restricted alternatives, without relying on the all-prime proof or any conjectural classification.

For the self-duality sentence on source page 9, use the following marking of $L(13)$ in degree 14:

$$\begin{aligned} r_0 &= (3\ 9)(4\ 10)(5\ 11)(6\ 12)(7\ 13)(8\ 14), \\ r_1 &= (1\ 4)(2\ 14)(3\ 9)(5\ 13)(6\ 11)(7\ 12), \\ r_2 &= (1\ 2)(3\ 7)(4\ 6)(8\ 14)(9\ 13)(10\ 12), \\ r_3 &= (1\ 2)(4\ 14)(5\ 13)(6\ 12)(7\ 11)(8\ 10). \end{aligned} \tag{6}$$

Both end groups have order 1092, and its type is $\{3, 7, 3\}$. The permutation

$$d = (1\ 3)(2\ 9)(4\ 7)(5\ 8)(6\ 12)(10\ 11)(13\ 14) \tag{7}$$

conjugates r_i to r_{3-i} for all four indices. Thus it normalizes the generated group and induces the required reversal automorphism. The resulting Class 1 maniplex is self-dual. This is an explicit

counterexample to the source's assertion about Class 1; the elementary face-count exclusion applies to Class 2 instead. A deliberately altered conjugator fails the four conjugacy identities in the checker.

The source's paragraph following Proposition 4.7 first correctly calls the local map of type $\{5, 3\}$ hemi-dodecahedral, then calls its facets hemi-icosahedral. Our statement preserves the actual local marking and uses the consistent hemi-dodecahedral name. None of these corrections is presented as an answer to an open conjecture, and no claim about unexamined editions is made.

9. Remaining source scope, formal coverage and cost

The unnumbered Suzuki observation in [LT, §5] uses a separate retained group fact: in the natural action of a simple Suzuki group, every involution fixes exactly one point, and commuting involutions have the same fixed point. In a rank-four string marking, r_0, r_2, r_3 would fix the same point, and then r_1, which commutes with r_3, would fix it too. The generated group would fix a point, contrary to the natural transitive action. The same argument applies to any higher rank, since every later generator commutes with r_0. This proves the stated rank bound from that retained Suzuki input. The general abstract centralizer version is already in the proof library; no new Suzuki classification is claimed here. The nonsimple small group Sz(2) cannot be generated by involutions at all, since its quotient C_4 cannot be so generated.

The PGL observations and final four-family conjecture of source §5 are kept separate from this PSL principal contract. In particular the literal claim there about every proper nonabelian subgroup of $P(q)$ omits its affine, dihedral and subfield subgroups. It cannot serve as a background classification. The companion work on that question retains the actual maximal-subgroup theorem instead.

The source rank restriction, Theorem 3.10, Proposition 4.2, Theorem 4.4, Lemma 4.5, Proposition 4.7 and Corollaries 4.6/4.8 are recovered in full. The local maps and 11-/57-cell classification remain attributed to their original authors and the reviewed common proof. The source matrices are matched by an explicit conjugacy, the two infinite families are proved for their full stated parameter ranges, and all necessary small fields are handled. The omitted symmetric-end family and the literal counterexamples are separate corrections. No complete census of all rank-four PSL markings is claimed or needed by the source contract.

The new Lean file proves three general statements: commutation extends from generators to the group; the first adjacent pair of a centre-free string marking does not commute; and the common-centralizer uniqueness condition forces rank at most four. These are the actual sggi bridge, not a theorem assuming IP. The characteristic-two bridge and the normal-form matrix calculations already have their separate, scoped certificates. The projective conjugacy/descent, Dickson classification, torus existence, finite universal orders, Dirichlet theorem and all positive construction claims remain written or computational proof obligations. They are not asserted to be covered by those three Lean declarations.

The full [E] umbrella and finite table are required mathematical inputs. The application adds the category bridge, complete source-specific constructions and explicit corrections. Its local and shared proof costs must be measured under the same convention as the source; importing a previously proved theorem does not erase its acquisition cost. This packet reports no compression ratio or chronological sawtooth, and awards no new principal credit before independent review.

References

[LT] D. Leemans and M. Toledo, *Maniplexes with automorphism group* $\mathrm{PSL}(2,q)$, author manuscript dated 15 May 2023, 11 pages. Bound primary: `sources/psl2_maniplex2023.pdf`.

[E] *Centralizers, facets and finite amalgams on the projective line*, frozen `phase238_topic_E_umbrella.md`, with the complete finite proof program `phase230_finite_check.g`. The subgroup inputs and older classifications retain the attributions given there.

[G] M. Giudici, *Maximal subgroups of almost simple groups with socle* $\mathrm{PSL}(2,q)$, arXiv:math/0703685v1 (23 March 2007), Theorem 2.2, physical and printed page 4. Bound primary: `sources/phase257_Giudici.pdf`. The maximal-subgroup classification is retained, not recovered as an additional source here.